



Wi-Sun (Image courtesy: <https://ytimg.com>)

Wi-Sun, short for 'wireless smart ubiquitous network,' is a leading global industry alliance promoting interoperable wireless standards based solutions for the IoT. It follows certain industrial standards. One among them is IEEE 802.15.4g, which is overall a mesh network that is believed to provide secure and cost-effective results. As Wi-Sun protocol provides good coverage in topographical environments and urban and rural areas with the lowest capital investments, it is believed to be an ideal replacement for star networking systems like LoRaWAN and NB-IoT.

Low-power dissipation makes Wi-Sun advantageous over all the wireless protocols because it has the lowest latency up to 0.02 seconds. Moreover, a high bandwidth up to 300kbps makes Wi-Sun compatible with multiple devices in longer range simultaneously. The mesh topology makes it robust and reliable of all.

Life-cycle of Wi-Sun based mesh networks, including the batteries involved, is said to be 15 years and longer.

"Overall, Wi-Sun is proposed as a future technology due to its lowest latency, high bandwidth and low power dissipation," says Niranjan. Above all, enterprises believe it to be a smarter solution economically.

To sum up

Although many upgrades have come up on a large scale that provide more benefits on economic front, there are some challenges before designers and enterprises.

Be it battery management for the Wi-Sun system or battery life-cycle, or miniaturisation of sensors, designers face issues from design to testing phase of a product. While packing more components in a smaller package, designers need to consistently focus on power management of the device too.

No matter how much the technology advances, cost is always going to be an effective measurement of the utility of a particular design upgrade. Thus, as we head towards artificial intelligence, virtual reality and IoT, cost will be a major concern. **EFY**

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